

HalluciCheck v2.0

AI Hallucination Verification System • Engineering Progress & Milestone Tracking

AUTHOR

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RELEASE VERSION

v2.0.0 (Production)

LIVE DEPLOYMENT

aihallucicheck.vercel.app

DATE

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HalluciCheck v2.0 — Engineering Progress & Milestone Tracking

Product: HalluciCheck v2.0 • AI Hallucination Verification System

Repository: <https://github.com/Niranchan67/aihallucicheck>

Live Production URL: <https://aihallucicheck.vercel.app>

Last Updated: September 21, 2026

Status: Production Ready (v2.0.0)

1. Executive Progress Summary

HalluciCheck has been transformed from an experimental heuristic script into an **autonomous, evidence-grounded AI hallucination verification system**. The platform decomposes arbitrary LLM-generated text into atomic propositions, executes parallel cross-examinations across 7 independent scholarly and authoritative registries, and calibrates a quantitative factual certainty score backed by verbatim source citations and deep multi-paragraph reasoning.

Metric	Status	Note
Atomic Claim Deconstruction	✓ 100%	Linguistic clause splitting + compound noun preservation
Multi-Source Retrieval	✓ 100%	OpenAlex, CrossRef, Europe PMC, ArXiv, Wikidata, DDG, Wikipedia
Entailment & Contradiction	✓ 100%	Strict relational logic, refutation probes, homonym filtering
Evidence Proofs & Citations	✓ 100%	Dataset matching, authority registry checks, live URL proofs
Frontend Workspace UI	✓ 100%	Production Light Theme, interactive sentence viewer, accessible
Cloud Deployment	✓ 100%	Live on Vercel Edge CDN + Unified Docker + Localtunnel

2. Chronological Milestones & Architecture Evolution

Milestone 1: Workspace Redesign & Visual Hierarchy (Production Light Mode)

- Design System:** Transitioned from dark-themed landing layouts to a focused **Single-View LLM Verification Workspace** strictly conforming to `AGENTS.md` guidelines.
- Palette & Typography:** Implemented high-contrast deep charcoal typography (`#0f172a`), muted secondary text (`#64748b`), and clean off-white background (`#f8fafc`).
- Color-Blind Accessibility:** Enforced triple redundant cues (color + icon + text label) across all statuses:
 - ✓ VERIFIED**: Soft Emerald (`bg-emerald-50 text-emerald-800 border-emerald-200`) + `CheckCircle2` icon.
 - ⚠ SUSPICIOUS**: Light Amber (`bg-amber-50 text-amber-800 border-amber-200`) + `AlertTriangle` icon.
 - ✗ HALLUCINATED**: Soft Rose (`bg-rose-50 text-rose-800 border-rose-200`) + `XCircle` icon.
- Interactive Highlighted Text Viewer:** Enabled sentence-by-sentence semantic highlighting with click-to-inspect popovers, keyboard accessibility (`tabIndex={0}` , `Enter` / `Space`), and `Esc` dismissal.

Milestone 2: Multi-Source Evidence Retrieval Engine

- **Independent Authority Repositories:** Eliminated single-source reliance on Wikipedia by building a federated concurrent retriever:
 1. **OpenAlex API:** Indexes 250M+ scholarly peer-reviewed works (Nature, Science, IEEE, ACM).
 2. **CrossRef DOI Registry:** Direct metadata validation for DOIs and academic works.
 3. **Europe PMC / PubMed Central:** Biomedical, medical, clinical, and life science literature.
 4. **ArXiv API:** Pre-print physics, mathematics, computer science, and AI archive.
 5. **Wikidata API:** Structured knowledge graph triples and relation definitions.
 6. **DuckDuckGo Web Search:** Real-time web index filtering for `.gov`, `.edu`, and primary sources.
 7. **Wikipedia REST API:** Retained strictly as secondary background context.
 - **Anti-Syndication:** Implemented root-domain duplicate suppression to prevent circular news syndicate mirrors from inflating confidence.
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Milestone 3: Atomic Proposition & Qualifier Decomposition

- **Fine-Grained Claim Decomposition:** Claims are analyzed beyond basic keyword matching into constituent propositions:
 - **Primary Relation:** (Subject, Relation, Object) core triple.
 - **Reason Qualifier:** Causal attribution ("for his discovery of...").
 - **Temporal Qualifier:** Chronological assertions ("in 1921").
 - **Location Qualifier:** Geographic assertions ("in Stockholm").
 - **Attribution Qualifier:** Institutional or personal credit ("awarded by the Royal Swedish Academy").
 - **Partial Support Rule:** If the main event occurred but secondary qualifiers (cause, date) are contradicted or fabricated, the assertion is flagged as **⚠ SUSPICIOUS** or **✗ HALLUCINATED** rather than falsely verified.
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Milestone 4: Strict Relational Entailment & Contradiction Resolution

- **Keyword Overlap Bug Elimination:** Fixed false-positive matches caused by lexical overlap:
 - Disambiguated non-geographic financial/economic terms (e.g., "capital expenditure", "social capital", "venture capital") from sovereign capital city assertions.
 - **Authoritative Contradiction Weighting:** When authoritative knowledge bases (Wikipedia, Wikidata, CrossRef) document an explicit refutation (e.g., Canberra is the capital of Australia, not Sydney), contradiction takes precedence over spurious open-web text.
 - **Targeted Contradiction Probing:** Automatically constructs counter-queries (e.g., "what is the capital of Australia") during retrieval to uncover refutations.
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Milestone 5: Linguistic Compound Sentence Splitting

- **Problem Resolved:** Compound sentences joined by conjunctions (e.g., "The capital of Australia is Sydney, and the human body consists of exactly 206 bones in an adult skeleton.") were previously merged into a single claim, causing true facts to mask false facts.
 - **Implementation:**
 - Implemented `_split_into_atomic_clauses()` in `claim_extractor.py`.
 - Analyzes coordinate and contrast clauses (`,\s+(?:and|but|whereas|while)` , `;` , em-dashes) using finite verb and subject dependency checks.
 - Preserves compound nouns intact (e.g., "Albert Einstein and Niels Bohr", "Water consists of hydrogen and oxygen" remain 1 claim).
 - Maintains exact character offsets (`start_index` , `end_index`) for sentence-level UI highlighting.
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Milestone 6: Deep Multi-Paragraph Reasoning & Proofs

- **Structured Rationale:** Every verified, suspicious, or hallucinated assertion generates a 3-paragraph factual assessment:
 1. *Executive Status & Calibration:* Factual certainty percentage, category tier, and core rationale.

- 2. *Empirical Ground Truth & Concordance*: Exact quotations from corroborating or refuting records.
- 3. *Authority & Methodology*: Authority weights of matched registries and justification for classification.
- **Structured Proof Tables**: Added `PropositionProof` , `AuthorityCheck` , and `EvidenceProof` data models passed from backend to frontend.

Milestone 7: Full-Stack Production Deployment & DevOps

- **Vercel Global Edge Deployment**:
 - React 18 + Vite frontend deployed to Vercel's global CDN at `https://aihallucicheck.vercel.app` .
 - Added `frontend/vercel.json` for client-side SPA routing.
 - Serverless-safe SQLite database fallback to `/tmp/hallucicheck.db` .
- **Unified Docker Container**:
 - Multi-stage `Dockerfile` compiling Vite frontend and serving on port 8000 via FastAPI.
 - `render.yaml` pre-configured for 24/7 cloud hosting on Render.com.
- **Instant Live Testing**: Localtunnel integration for immediate remote testing from any machine.

3. Component Verification Status Matrix

Component	File Path	Status	Verification Check
Claim Extractor	<code>backend/services/claim_extractor.py</code>	✔ Complete	Splits coordinate clauses, preserves compound nouns, tracks character offsets
Claim Analyzer	<code>backend/services/claim_analyzer.py</code>	✔ Complete	Decomposes claims into 11 internal taxonomy categories and atomic propositions
Query Generator	<code>backend/services/independent_verifier.py</code>	✔ Complete	Generates neutral primary queries, subclaim queries, and contradiction probes
Multi-Source Retriever	<code>backend/services/multi_source_retriever.py</code>	✔ Complete	Concurrently queries OpenAlex, CrossRef, Europe PMC, ArXiv, Wikidata, DDG, Wiki
Source Validator	<code>backend/services/source_validator.py</code>	✔ Complete	Classifies source authority tiers (1.00 to 0.45) and evaluates semantic relevance
URL Validator	<code>backend/services/url_validator.py</code>	✔ Complete	Probes HTTP reachability, follows redirects, canonicalizes URLs
Entailment Engine	<code>backend/services/entailment_engine.py</code>	✔ Complete	Evaluates proposition-level entailment, resolves contradictions, prevents homonyms
Confidence Scorer	<code>backend/services/confidence_scorer.py</code>	✔ Complete	Computes clinical certainty score (0-100%) and tabular risk distributions
Citation Validator	<code>backend/services/citation_validator.py</code>	✔ Complete	Resolves DOIs via doi.org and verifies academic references via CrossRef
FastAPI Core	<code>backend/main.py</code>	✔ Complete	REST endpoints (<code>/api/verify</code> , <code>/api/health</code> , <code>/api/verifications</code>), CORS enabled
Interactive Results	<code>frontend/src/components/LightResults.tsx</code>	✔ Complete	Color-coded sentence spans, modal inspection cards, proofs table, citations
Analysis Console	<code>frontend/src/components/AnalysisConsole.tsx</code>	✔ Complete	Paste-friendly textarea, word counters, preset chips, keyboard shortcuts (<code>Ctrl+Enter</code>)

4. Empirical Test Results & Benchmarks

Test Benchmark 1: Multi-Part Compound Coordinate Sentence

- **Input**:

"The capital of Australia is Sydney, and the human body consists of exactly 206 bones in an adult skeleton."

- **Outcome:**
 - **Claim 1:** "The capital of Australia is Sydney." → **X HALLUCINATED (14.0% Certainty)**
 - *Contradiction Details:* Refuted by Wikipedia/Official records: Canberra is the capital of Australia; Sydney is the state capital of New South Wales.
 - **Claim 2:** "The human body consists of exactly 206 bones in an adult skeleton." → **✓ VERIFIED (88.0% Certainty)**
 - *Corroboration:* Directly substantiated by anatomical records.
 - **Overall Score:** 50.0% (**MODERATE RISK**).
 - **Visual Viewer:** Sentence 1 rendered in **Soft Red**; Sentence 2 rendered in **Soft Green**.

Test Benchmark 2: Compound Noun Preservation

- **Input:** "Albert Einstein and Niels Bohr contributed to the foundations of quantum mechanics."
- **Outcome:** Correctly kept as **1 atomic claim** (compound subjects joined by "and" are not split).

Test Benchmark 3: Causal Reason Qualifier Verification

- **Input:** "Albert Einstein won the Nobel Prize in Physics in 1921 for his discovery of the law of the photoelectric effect."
- **Outcome:** Evaluated across 3 propositions (primary award, 1921 date, photoelectric effect reason). Ground-truth citation matches official Nobel citation.

5. Live Environments & Deployment Endpoints

Environment	Host / Platform	URL / Command	Notes
Production Frontend	Vercel Edge CDN	https://aihallucicheck.vercel.app	Global CDN, instant loading, SPA routing
GitHub Source	GitHub	https://github.com/Niranchan67/aihallucicheck	Complete source tree, Dockerfile, configs
Unified Local Server	Local Uvicorn	<code>http://127.0.0.1:8000</code>	Single-port fullstack backend + React build
Local Dev Frontend	Vite Dev Server	<code>http://localhost:5173</code>	Hot Module Replacement (HMR) active
Remote Tunnel	Localtunnel	<code>https://some-peaches-cough.local.lt</code>	Remote access tunnel active for mobile testing

6. Future Roadmap

1. **PDF / Document Drag-and-Drop:** Ingestion of multi-page PDF research papers and automatic claim batching.
2. **Claim Clustering Graph:** Interactive force-directed network graph showing connections between related claims and shared evidence clusters.
3. **Automated Red-Teaming Benchmark Suite:** Continuous automated evaluation against standardized hallucination datasets (HaluEval, FactCC, TruthfulQA).